

Magnetic Susceptibility

A Physical Chemistry Laboratory Experiment

It is increasingly possible to teach the interpretation of the properties of substances in terms of their molecular and electronic structure in the undergraduate physical and inorganic chemistry courses. One interesting field of endeavor, which has become very important recently, is that of the magnetic behavior of materials.

The goal of a properly designed magnetic susceptibility laboratory exercise is to show the relation between macroscopically measured magnetic properties and electronic structure. The purpose of this article is to describe an experiment designed to reach this goal that has been run successfully by undergraduate students. A Gouy Balance is used to measure magnetic susceptibilities of various *solid* materials as a function of field strength. Such an experiment has been described previously (1-3). However, the present experiment differs both in object and in experimental detail. In particular, solid samples are chosen which give very large deflections in the magnetic field thus necessitating less sensitive apparatus; deflections observed are on the order of one-half gram. From these deflections, one infers the presence of any possible ferromagnetic impurities, the magnetic susceptibilities, and the number of unpaired electrons. The results are interpreted in terms of the electron configuration of the compound.

It is hoped that the write-up² compels the student to make extensive use of the library for interpretation of his results, and time is allowed for this. The reactions of the students who do the experiment is that they have to work hard and that they are satisfied with their accomplishments.

Experiment

The Gouy Balance. Measurements of magnetic susceptibility are made on a Gouy balance. As shown in the figure, the sample is placed in a long tube of uniform cross section which is suspended vertically from an ordinary chainomatic balance between the two pole faces of an electromagnet. The bottom (or the top, in some cases) of the sample is leveled with the center of the pole faces. The sample is thus situated in a magnetic field which is inhomogeneous along the vertical axis, varying from H_0 at the bottom to H at the top

When the sample is suspended from an analytical balance, the apparent increase in mass Δm due to the magnetic field is

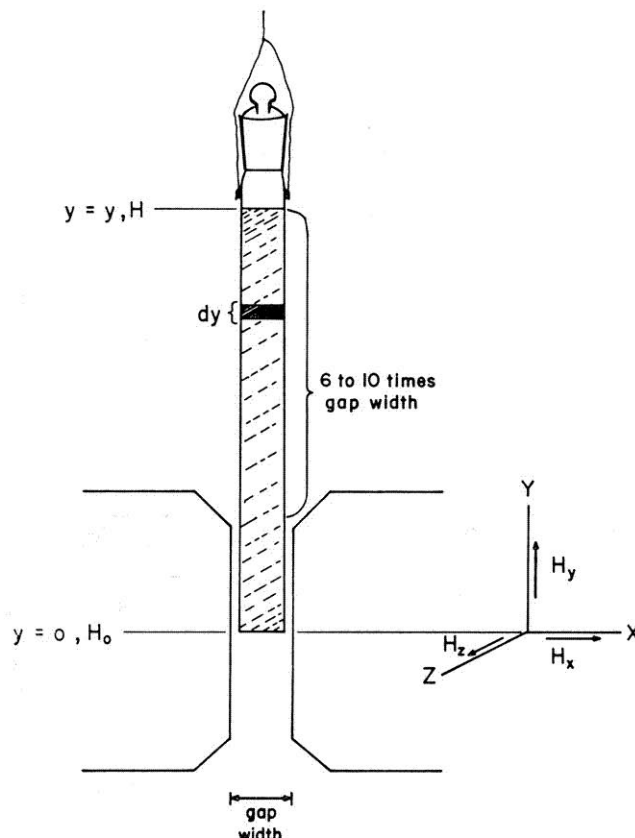
$$\text{force} = (\Delta m)g = \frac{1}{2}(K_2 - K_1)AH_0^2$$

where g is the gravitational acceleration (5); K_2 is the volume susceptibility of sample and K_1 that of surrounding medium (usually air) ($K(\text{air})$ is usually negligible compared to $K(\text{sample})$); and A is the area of the sample tube.

Measurements. Measurements are made on paramagnetic and ferromagnetic *solids* which have relatively large susceptibilities.

In general the sample should be ground to a fine powder to avoid anomalies due to anisotropy in the crystal and to facilitate *uniform packing*, which is essential. Complete uniformity of crystal size is difficult to achieve, and usually unnecessary if the crystals are small. Crystals of less than 0.1 mm are quite readily obtained using a mortar and pestle, but in some cases crystals of larger size will give satisfactory results.

Mohr's salt (iron (II)) ammonium sulfate hexahydrate— $\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ which has a large paramagnetic moment and is stable in air is used as a standard. This compound has the notable advantage of being commercially available as a certified standard for use in quantitative analysis.



This drawing shows a uniformly packed sample of isotropic material in an inhomogeneous magnetic field. If the sample is paramagnetic, it will experience a downward force when the magnetic field is turned on. Representative dimensions are sample tube, 14-18 mm od $\times$ 150 mm and magnet, 100 mm core, 75 mm pole piece, 22 mm gap width.

¹ Address reprint requests to Professor Stafford.

² The authors of this paper have written a 35 page "student hand-out" booklet which gives a complete theoretical introduction as well as experimental detail. It is available at cost from Professor Stafford.

Measurements are made on several materials with interesting electron configurations and which are stable in air. Suitable solids are iron(III) ammonium sulfate, copper(II) sulfate pentahydrate, anhydrous copper(II) sulfate, copper(II) oxide, gadolinium sulfate octahydrate, 2,2-diphenyl-1-picrylhydrazyl (used as a standard for ESR work), and ordinary aluminum oxide which usually contains ferromagnetic impurities.

Treatment of Data

The gram susceptibility of Mohr's salt is given by

$$\chi = \frac{9500 \times 10^{-6}}{(T + 1)}$$

where T is the absolute temperature. We have (1-3, 5)

$$H_0 = \sqrt{\frac{2g\Delta m}{\chi Ad}}$$

where d is the "effective" density for the particular sample used. The magnetic field strength H_0 is calculated and plotted as a function of the magnet energizing current.

For the other solids, χ is calculated from the above equation, and its constancy noted as a function of field strength. Finally μ_{eff} , the effective number of Bohr magnetons, and n , the number of unpaired electrons per transition element atom, may be calculated (1-3, 5).

Results

In the course of two 3-hour laboratory periods a pair of students is able to make measurements on four compounds at each of four field strengths ranging from 0 to 7000 gauss inclusive. A set of student results taken at one field strength is given in the table.

An excellent compilation of magnetic susceptibilities has been made by Foëx (6). Some magnetic susceptibilities are recorded in the "Handbook of Physics and Chemistry." With care an over-all accuracy within several per cent of the true value may be obtained for paramagnetic substances.

Impure substances may give results that are greatly in error, particularly if the impurity is ferromagnetic. Trace ferromagnetic impurities may be detected by an apparent dependence of the susceptibility on the field strength.

Discussion

This experiment illustrates the use of magnetic measurements to determine the number of unpaired electrons in a compound. On the basis of the number of unpaired electrons found, possible electronic structures are drawn for each compound showing both possible and excluded structures, if any.

Magnetic Susceptibility Measurements

Substance	Δm^a (grams)	Magnetic susceptibility $\times 10^6$	
		obs.	lit.
FeSO ₄ (NH ₄) ₂ SO ₄ ·6H ₂ O	0.8739	(standard)	31.6
CuSO ₄ ·5H ₂ O	0.1924	5.6	5.9
CuSO ₄ (anhydrous)	0.2751	8.5	9.6
DPPH ^b	0.0248	2.8	^c
Glass sample cell	-0.006	...	...

^a Sample size, about 30 g of solid, except for DPPH, 10 g. Magnetic field, ca. 5000 gauss.

^b 2,2-Diphenyl-1-picrylhydrazyl (NO₂)₃C₆H₂NN(C₆H₅)₂.

^c See reference (8), Table 1.

It will be found that cupric oxide exhibits a considerably lower paramagnetic susceptibility than expected from a substance with one unpaired electron. This is apparently due to exchange interactions (7). If paramagnetic ions are in close proximity in a crystal, interactions between the unpaired electrons of adjacent ions may occur which may either increase or decrease the observed magnetic susceptibility. It is because of this exchange interaction that iron and some of its compounds have such enormous susceptibilities (ferromagnetism), whereas the dispersion of Cr₂O₃ in Al₂O₃ results in an increase in the magnetic susceptibility of Cr₂O₃ (5). The exchange effect is diminished in "magnetically dilute" materials such as Fe(NH₄)-(SO₄)₂·12H₂O in which the ferric ions are quite widely separated due to the other diamagnetic species present in the crystal structure.

Although the transition elements form the most common paramagnetic compounds, other substances may contain unpaired electrons. Certain stable organic free radicals are known and have been studied (8).

Other Applications

In recent years magnetic measurements have been utilized in a wide variety of chemical studies in addition to those in coordination chemistry (9). In the common type of dry cell, manganese dioxide (the only paramagnetic material in the cell) acts in a secondary manner as a depolarizer; in this process it is reduced and the number of unpaired electrons in the manganese increases. Magnetic measurements have been used to measure the extent of the reduction and have further indicated that the manganese is reduced during the discharge of the cell, not during the recovery period (10).

Another interesting general application of magnetic measurements is that of magnetic titrations. An example is the work of Coryell, *et al.* with hemoglobin (11). There are many other applications. One recent development worthy of special note is the study of bonding during adsorption on catalyst surfaces (12).

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